CCC-C-460
SEPTEMBER 1, 1960
SUPERSEDING
Fed. Spec. CCC–F–456a
April 8, 1960

FEDERAL SPECIFICATION

CLOTH, COTTON, FLANNEL, (HEAVY, FOR TABLE FELTS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of cotton flannel.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.**—The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein.

Federal Specifications:

CCC–T–191—Textile Test Methods.
PPP–P–51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4 — Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspecton by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Textile Fiber Products Identification Act.

(Copies may be obtained, free of charge, from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Material.**

3.1.1 *Yarn.* — The yarn shall be evenly spun from clean, carded, and drawn cotton.

3.2 **Physical requirements.**—The finished flannel shall conform to the requirements shown in table I when tested as specified in 4.3.

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TABLE I.—*Physical requirements*

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength, grab method (minimum)	
	Warp	Filling	Warp	Filling
Ounces			Pounds	Pounds
12.5	66	42	60	30

3.2.1 *Width.*—The overall width shall be 48, 54, 64, or 72 inches, as specified (see 6.2). A minus tolerance of ½ inch and any plus tolerance will be permitted.

3.2.2 *Weave.*—The flannel shall have a filling reversible (or filling back) weave, i.e., one warp and two filling systems, weaving as a 3 down, 1 up straight or broken twill when viewed on either the face or the back.

3.3 Finish.—The flannel shall be unbleached and shall have a full nap on both sides.

3.4 Length. — Unless otherwise specified (see 6.2), the flannel shall be put up in rolls of not less than 20 continuous yards for 48- and 54-inch widths and not less than 30 continuous yards for 64- and 72-inch widths.

3.5 Fiber identification. — Each piece of felt shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act.

3.6 Workmanship. — The finished flannel shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. —Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. — Except when otherwise specified by Civil Government agencies, inspection shall be in accordance with Military Standard MIL–STD–105 except where otherwise indicated.

4.2.1 *Examination of the end item.*—Defects found during the examination shall be classified in accordance with list shown in 4.2.1.1, 4.2.1.2, 4.2.1.3, and 4.2.1.4.

4.2.1.1 *Yard-by-yard examination.* — The required yardage shall be examined on one side only and defects classified as listed. The side examined shall be alternated on every other piece. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product shall be 1 linear yard. The sample size shall be based upon inspection level II of Military Standard MIL–STD–105. The sample size and the lot size shall be expressed in units of 1 yard each. The number of rolls from which the sample is to be selected shall be in accordance with table II of this specification. An approximately equal number of yards shall be examined in each roll of the sample. The acceptable quality levels (AQL's) shall be as follows:

Widths	*AQL's*
48 inch and 54 inch -----	1.5 major — 6.5 totals
64 inch and 72 inch -----	2.5 major — 10.0 totals

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Glossary No. of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
	General defects (defect not restricted to a direction and defects applicable to both warp and filling)		
90, 91, 92	Any hole, cut or tear, three or more yarns ruptured in adjoining points	X	
78	Any smash	X	
79	Any abrasion mark or chafe resulting in a thin or weak place, clearly noticeable[1]	X	
74, 75, 76	Any spot, stain or streak clearly noticeable[1] more than ½ inch in combined length and width directions		X
11, 33	Two or more contiguous yarns, broken or missing for 3 inches or more in length	X	
	Two or more contiguous yarns broken or missing for less than 3 inches in length		X
	Single yarn broken 9 inches or more		X
5, 27	Any slub, slug or bunch, three or more times the thickness of the normal yarn diameter and 1½ inches or more in length	X	
	Any slub, slug or bunch, three or more times the thickness of the normal yarn diameter and more than ¾ inch but less than 1½ inches in length		X
	Any slub, slug or bunch, thicker than the normal yarn diameter but less than three times the thickness and 1½ inches or more in length		X
13, 34	Multiple floats or skips ¾ inch or more in combined warp and filling directions	X	
	Multiple floats or skips less than ¾ inch in combined warp and filling directions		X
	Single, more than 4 inches in length		X
15, 37	Tight yarn causing waviness in adjoining area		X
	Warpwise defects: (Defects restricted to the warpwise direction)		
35	Hitchbacks, clearly noticeable[1], distorted area two inches or more in combined direction		X
	Fillingwise defects: (Defects restricted to the fillingwise direction)		
2, 9	Fine or heavy filling bar, light or heavy place, clearly noticeable[1]	X	
4, 10	Double pick or mispick, clearly noticeable[1]		X
3	Jerked in filling more than 1/16 inch in thickness and 4 inches or more in length	X	
	Jerked in filling more than 1/16 inch in thickness and more than 2 inches but less than 4 inches in length		X
12	Crack opening at widest point ⅛ inch or more	X	
	Crack, opening at widest point less than ⅛ inch		X
22	Finger mark or pressure mark, clearly noticeable[1] and extending over 1 inch		X
	Selvage defects: (Defects restricted to the selvage)		
49, 50	Tight or slack selvage, resulting in pronounced slackness or waviness in adjoining cloth	X	

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Glossary No. of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
	Selvage defects:—Cont.		
51	Scalloped selvage, deepest indentation more than ½ inch		X
55, 56	Cut, broken or torn selvage	X	
	Finishing defects: (Defects restricted to the finishing operations)		
57	Finishing mark, clearly noticeable[1]		X
59, 61	Crease or wrinkle, hard embedded		X
69, 70	Dye stain or streak, clearly noticeable[1]		X

[1] Clearly noticeable at normal inspection distance approximately 3 feet.

4.2.1.2 *Overall examination.*—Each defect listed shall be counted not more than once in each piece examined. The unit of product shall be one piece. The number of pieces selected for yard-by-yard examination shall comprise the sample for this examination and the number of defects acceptable shall be as specified in table II of this specification.

Defects

Uneven nap—either side
Overall uncleanness
Rancid or other objectionable odor
Rolled, folded, beaded, wavy or tight selvage over more than ½ the length of the piece.
Width—less than minimum specified by more than ½ inch

4.2.1.3 *Examination for length.*

4.2.1.3.1 *Individual pieces.*—The piece shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards below the gross length marked on the ticket shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the acceptance number shall be in accordance with table II.

TABLE II.

Lot size in yards	Sample size in rolls	Maximum No. of defects acceptable in sample
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 to 550,000	25	1

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.1.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total of the gross lengths marked on the tickets.

4.2.1.4 *Examination of preparation for delivery requirements.* — An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with

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the following list. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, or improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability. Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

4.3 Testing of the end item.—The methods of testing specified in Federal Specification CCC-T-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be two yards full width. The lot will be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method
Fiber identification	3.1.1	1200
Yarns per inch:		
Warp	Table I	5050
Filling		
Weave:	3.2.2	Visual
Breaking strength:		
Warp	Table I	5100
Filling		
Weight, ounces/square yard	Table I	5041

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for Civil agencies.)

5.1 Packaging.—Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.*—The cloth put up on rolls shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing.—Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.*—The cloth, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking.—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use.—The cloth covered by this specification is intended for use as table felts.

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6.2 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Width required (see 3.2.1).

(*c*) Length of rolls if other than specified (see 3.4).

(*d*) Use of Military Standard MIL–STD–105 (see 4.2).

(*e*) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it should be noted that various options, choices and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of cloth.

6.3 Level B packaging and packing should be used by ordering agencies only when shipments are known to be made to a holding Supply Depot or Warehouse from which partial shipments will be made to ultimate consumers.

6.4 **Transportation description.** — The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods not otherwise indexed by name in the original piece.
Carload minimum weight — Any quantity.

Motor:

Cotton piece goods not otherwise indexed in the original piece.
Motor volume minimum weight —Any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—MQ
Navy—S
Air Force—WADC.

Copies of this specification may be purchased for

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